ENVIRONMENTAL ANALYSIS (NWTC)

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Environmental Chemical Analysis (NWTC)

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CHAPTER OVERVIEW

1: General Lab Safety

1.1: Laboratory Area

General Safety Practices

Lab Expectations

Food and beverages are prohibited within the lab space. Pathogens or toxic chemicals could transfer to your food or beverage, leading to illness or even death.

The environmental lab is divided into two specific areas. The classroom half is used for non-lab activities. This area accommodates backpacks, jackets, and learning equipment such as laptops and notebooks. In the classroom area, contamination risk is low but not zero. The lab half is used for experimentation, and the contamination risk is high. Students should take extra caution in the lab area. Lockers are available in the hallway for student use.

Lab Safety

Prior to beginning any experiment, students will review safety information. Safety information will include proper procedures, chemical hazards, biological hazards, and physical hazards. Students need to locate important safety equipment such as the first aid kit, emergency eyewash, emergency shower, fire extinguisher, and chemical spill kits. Students also need to locate the nearest exit in case of an emergency.

1.2: Personal Protective Equipment

While the lab does minimize risk using engineering and administrative controls, personal protective equipment is required to mitigate the impact of hazards should other controls fail. Students are expected to wear the following during lab experiments where hazards are present:

- Long pants
- Closed-toed shoes
- Lab coat
- Safety glasses or goggles
- Gloves

1.3: Chemical Handling

Safety data sheets are available for all chemicals used in the lab. Students, when preparing for lab, must review the safety data for any chemicals being used. Below are key points for the proper storage of laboratory chemicals:

- Chemicals must be stored in a cool, dry location out of direct sunlight
- Chemical storage in a cabinet is recommended
- Liquid chemicals should be stored below human face level to avoid splattering in case of bottle spillage or breakage
- Chemicals must be dated upon receipt, and the expiration date must be recorded
- Acids should be carefully stored, separate from other chemicals, and below eye level
- Chemicals should never be stored in unmarked containers

Other Chemical Handling Notes

- Always pour acid into water when diluting and swirl or mix to dissipate heat
- The pH of liquids poured down the drain must be between 4 and 1

CHAPTER OVERVIEW

2: Laboratory Equipment

2.1: Glassware

Basic Glassware

Glassware	Description and Use
Volumetric flask	Bulb-shaped with a long stem. This is used to measure only one specified volume. It is best to use when making standards for calibration.
Beakers	Cylinder-shaped with a large open top. While these have volume measurements, they are not accurate enough to use for measuring. These are best used for holding liq
Erlenmeyer flask	They have wide bottoms and a narrow top. Like beakers, these are not accurate enough to use for measurement. These are best used for holding liquid that needs to b mixed.
Graduated cylinders	Tube-shaped glassware with many measurement increments. These are used for measuring liquids. Choose a size close to the volume you need to measure. Be sure bottom of the meniscus.
Burette	Similar to a graduated cylinder, but liquid drains out the bottom from a valve. High accuracy and is often used for titrations.

Pipets

Volumetric pipets are classified as "TD" (To Deliver; the new international designation is "EX") and should never be blown out. These pipets are calibrated to deliver one specific volume, and the calibration already takes into consideration the small amount of volume that may remain in the tip. Volumetric pipets may have a colored square at the top of the pipet, which is a manufacturer’s mark indicating the volume of the pipet.

Serological pipets can be used to measure various volumes from a single pipet due to their volume gradations. There are two kinds of serological pipets, "blow out" and "drain out". Serological pipets from which the volume in the tip should be "blown out" typically are graduated right up to include the entire tip. These pipets are usually identified by the presence of two thin colored or etched (frosted) rings at the top of the pipet.

Serological or Mohr pipets, which should be "drained out," typically have their graduated markings stop prior to the tip. The volume should not be drained below the last marking. These pipets are usually identified by the presence of a color-coded rectangle or square (but no frosted or colored rings) at the top of the pipet.

2.2: Analytical Probes

pH Probe

Water has a wide range of low pH acids and high pH bases. In most instances (drinking water, surface waters), we want water to be close to neutral (pH = 7) for both safety and taste. Water treatment processes and chemical dosing might be optimized in different pH ranges, prompting water treatment operators to both maintain and adjust pH levels based on treatment. Levels of pH are determined by the amount of hydrogen ions (H^+), causing acidity, or hydroxide ions (OH^-), causing basicity.

The pH probes are stored in a solution and must be wrapped to keep the solution from evaporating. The probe uses the internal electrolyte solution to calibrate and measure pH. Calibration of the pH probe uses three different pH buffers.

Conductivity Probe

Water conductivity is a measure of the dissolved mineral salts in water. High levels of mineral salts will give high conductivity readings. Calibration of the conductivity probe uses two different conductivity standards. Standards chosen for calibration should be one lower than your anticipated reading and one higher than your anticipated reading.

2.3: Light Based Meters

Turbidimeter

Turbidity is an optical measurement of water quality. Within a turbidimeter, light will shine through the sample, particles suspended in the water will cause the light to be scattered, and a meter will measure the light able to pass through the sample. Turbidimeters are calibrated with standard solutions before measuring samples.

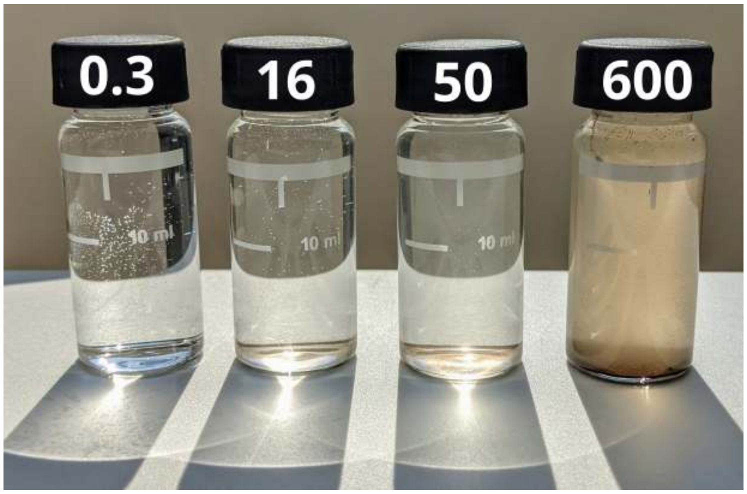

Figure 2.3.1: Sample containers showing a range of turbidity.

Spectrophotometer

A spectrophotometer is a device used to measure light intensity through a liquid. When using the spectrophotometer, the following happens:

1. A light source shines through the sample
2. Some of that light gets absorbed by the sample
3. A sensor measures how much light has traveled through the sample and, therefore, how much light was absorbed by the sample
4. The machine provides a numerical output for the light absorbance.

When using a spectrophotometer in the lab, we will compare unknown samples to a clearly defined set of standard dilutions. The standard dilutions can be plotted graphically, providing a baseline that will calculate our unknown sample. Spectrophotometers are highly accurate machines and are often used for permit compliance testing or for sensitive chemical measurements, like with total phosphorus.

Colorimeter

Colorimeters are similar to spectrophotometers but are less sensitive and therefore typically more affordable. Colorimeters use internal red, green, and blue filters to measure levels of color differences in chemically altered samples. Colorimeters can be used to monitor a variety of water treatment chemicals, including fluoride, chlorine, and phosphate.

CHAPTER OVERVIEW

3: Laboratory Foundations

3.1: Chemistry Review

The Periodic Table of Elements

The periodic table of elements is a map of all known chemical elements organized by their atomic number, electron configuration, and chemical properties. **Groups** refer to columns in the table, and **periods** refer to the rows. Based on an element's position in the table, scientists can predict how an element will interact with other types of elements.

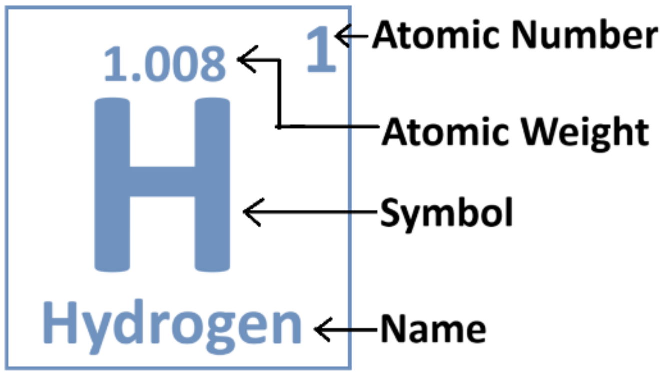

Figure 3.1.1: Hydrogen's entry in the periodic table of elements.

Stoichiometry

Stoichiometry quantifies the relationships between reactants and products in a chemical reaction. Based on a balanced chemical equation, stoichiometry can determine the amounts of substances consumed and produced in a reaction. Substances can be compared by mole ratios and mass ratios.

Balancing chemical equations is a necessary first step. To achieve a balanced equation, the quantity of specific elements on the reactants side (left side) needs to equal the quantity of specific elements on the products side (right side). Begin by writing out the chemical equation, then balance the metals, non-metals, and oxygen.

Example 3.1.1

Write the balanced chemical equation for when magnesium metal undergoes combustion to produce magnesium oxide.

Solution

Write out the reactants and products of the equation

$$Mg + O_2 \text{ --> } MgO$$

Count the number of each atom on the reactants and products sides of the equation

Reactants		Products	
Mg	1	Mg	2
O	2	O	2

Use whole-number coefficients to balance each side

- Since there are two oxygen on the reactants, add a coefficient of 2 for the products

$$Mg + O_2 \text{ --> } 2\,MgO$$

- Now the oxygen is balanced, but the products side has two magnesium, while the reactants side has one.
- Add a coefficient of 2 before magnesium on the reactants side

$$2\ Mg\ +\ O_2\ \text{-->}\ 2\ MgO$$

- Recount the number of each atom on the reactants and products sides of the equation. They should be equal

Reactants		Products	
Mg	2	Mg	2
O	2	O	2

You have found your balanced equation

$$2\ Mg\ +\ O_2\ \text{-->}\ 2\ MgO$$

Electronegativity

Electronegativity is a measure of an atom's ability to attract and hold onto electrons within a chemical bond. It plays a crucial role in determining the nature of chemical bonds and the behavior of molecules. Here's how it relates to the periodic table:

1. **Trend Across Periods**: Electronegativity generally increases as you move from left to right across a period. This is because atoms have more protons in their nuclei, which increases their ability to attract electrons.
2. **Trend Down Groups**: Electronegativity decreases as you move down a group. This is due to the increasing distance between the nucleus and the valence electrons, which reduces the nucleus's pull on the electrons.

Electronegativity plays a role in chemical bond types. **Ionic bonds** occur when there is a large difference in electronegativity. This difference leads to the transfer of electrons from one element to another and typically happens between metals and non-metals (e.g., NaCl). **Covalent bonds** occur when there is a small difference in electronegativity. In covalent bonds, each atom contributes electrons to the bond, which exist in overlapping orbitals. Water (H_2O) is a classic example of a compound with covalent bonds. In a water molecule, each hydrogen atom shares one electron with the oxygen atom.

pH

pH is a scale measuring hydrogen ion concentrations in water. The scale ranges from 0 to 14, with low numbers indicating acidity, 7 representing neutral, and high numbers indicating basicity. The pH scale is logarithmic, so each whole-number change represents a factor of ten in hydrogen ion concentration.

Many chemical reactions are driven by the solution's pH level. The rate of reaction and the equilibrium point can change as pH increases or decreases. In water, changes in pH can modify nutrient availability. While this can negatively impact the environment, water treatment facilities use pH to their advantage when treating water. Adjusting the pH can optimize water treatment processes and decrease the overall chemical needs.

3.2: Weighing

Analytical Balance

Analytical balances are most often found in a laboratory where extreme sensitivity (0.1 mg) is needed for the weighing of items. Analytical balances usually measure mass and range in capacity from 0.1 mg to 150 grams. They are typically used for the suspended solids (TSS) testing in the wastewater laboratory, where weight changes of 0.1 mg must be measured.

General Purpose or Top-Loading Balances

General-purpose or top-loading balances are also used primarily in a laboratory setting. They usually can measure objects weighing around 150-300 g. They offer less readability and accuracy than an analytical balance, but allow measurements to be made quickly, thus making it a more convenient choice when exact measurements are not needed. Top-loaders are also more economical than analytical balances and are suitable for weighing chemicals when preparing reagents.

Balance Performance

For best performance, balances need to be set up correctly. The following is a checklist for proper balance use:

- The electronic balance is turned on and allowed a warm-up time
- Balance is set on a sturdy foundation protected from vibration
- Balance is level
- Temperature and humidity are relatively constant, and samples are at room temperature
- The balance and the pan are clean, and the material is weighed in a container
- If used, sliding doors must be closed when making measurements
- Balances can be calibrated using ASTM Type 1 weights. These weights should be stored carefully and only handled with gloves

3.3: Accuracy, Precision, and Error

Quality Control Standards

Quality control standards are samples prepared with a known amount of analyte. The purpose of these standards is to determine if the laboratory methods and equipment are capable of making accurate and precise measurements. These standards are important indicators of quality and provide instantaneous feedback on method and equipment performance.

Accuracy and Bias

Accuracy is the degree of closeness to the actual value. The goal of sample testing is to find the sample's true measured value for water quality parameters.

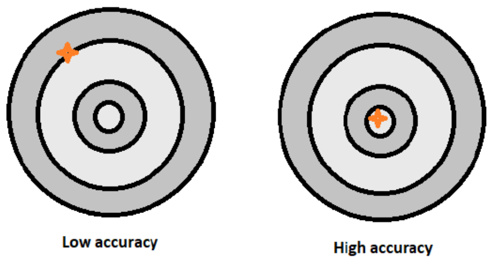

Figure 3.3.1: The difference between low and high accuracy.

Bias is the systematic or persistent error in an analysis that results in the expected sample measurement being consistently different than the sample's true value. Bias results from a flaw integral to the testing system. Systematic bias will lead to systematic errors that skew high or low, as opposed to random errors, which generally cancel each other out.

> An incorrectly calibrated home thermostat may consistently read several degrees hotter or colder than the actual temperature. This is an example of bias that commonly leads to systematic errors.

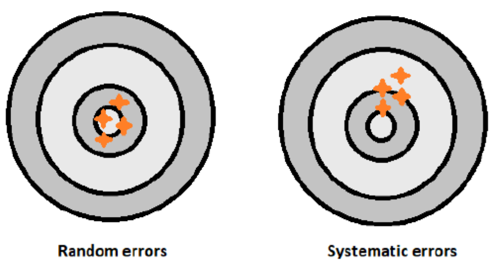

Figure 3.3.2: The difference between random and systematic errors.

Precision

Precision is a measure of how closely multiple determinations performed on the same sample will agree with each other. Performing a test on a sample many times and measuring the same result each time is an example of precision. This does not mean, however, that the measured result is accurate. Systemic bias does not affect precision like it does accuracy.

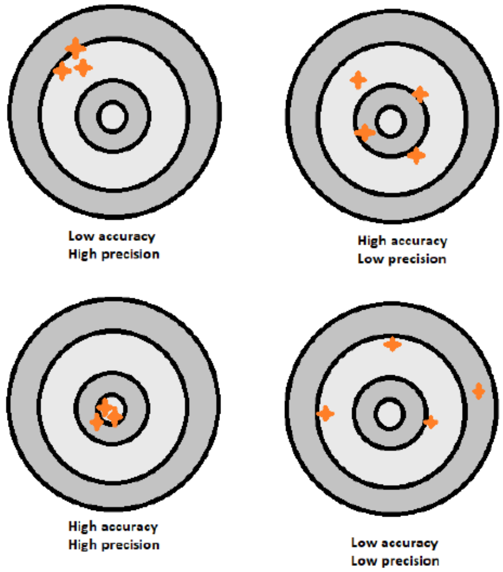

Figure 3.3.3: Illustration of combinations of accuracy and precision.

Error

Error is the difference between the measurement and the true value. Error is expected in many experiments, even if it is a very small amount. To account for these small errors, uncertainty values are set to indicate a range of values, based on the measurement, that the true value will be in.

3.4: Quality Assurance and Quality Control

Quality Assurance

Quality assurance is an integrated system of management involving planning, implementation, documentation, assessment, reporting, and quality improvement to ensure that a process or item meets the needs of the experiment.

This includes the preparation of standard test methods, maintaining the laboratory at a constant temperature and humidity, sending equipment in for factory calibration, and maintaining chemicals within their recommended shelf life. Quality assurance should be regularly reviewed, and adjustments or corrective actions should be made as needed to achieve quality goals.

Quality Control

Quality control is the overall system of technical activities that measures the attributes and performance of a process against defined standards to verify that it meets the stated requirements. This includes operational techniques and activities used to fulfill requirements for quality.

Quality control measures can include calibrating instruments, measuring blanks or control samples, and providing training for lab technicians.

> **Note**
>
> Definitions referenced from the Forum on Environmental Measurements Glossary, Environmental Protection Agency
>
> Environmental Measurements and Modeling | US EPA

CHAPTER OVERVIEW

4: Soil Properties

4.1: Soil Health

Soil Health

Soil health is defined as the continued capacity of soil to function as a vital living ecosystem that sustains plants, animals, and humans. Healthy soil gives us clean air and water, bountiful crops and forests, productive grazing lands, diverse wildlife, and beautiful landscapes. Soil does all this by performing five essential functions:

- **Regulating water**
  Soil helps control where rain, snowmelt, and irrigation water go. Water flows over the land or into and through the soil.
- **Sustaining plant and animal life**
  The diversity and productivity of living things depend on soil.
- **Filtering and buffering potential pollutants**
  The minerals and microbes in soil are responsible for filtering, buffering, degrading, immobilizing, and detoxifying organic and inorganic materials, including industrial and municipal by-products and atmospheric deposits.
- **Cycling nutrients**
  Carbon, nitrogen, phosphorus, and many other nutrients are stored, transformed, and cycled in the soil.
- **Providing physical stability and support**
  Soil structure provides a medium for plant roots. Soils also provide support for human structures and protection for archeological treasures.

Reference

- Natural Resources Conservation Service (usda.gov)

4.2: Dynamic Cone Penetrometer

Dynamic Cone Penetrometer

A dynamic cone penetrometer (DCP) is a tool used for evaluating the strength of soils on-site. It also helps with monitoring the condition of granular layers and subgrade soils in pavement sections over time. It can be used to determine the right solutions for the sites, especially when soft soils are involved.

The DCP uses a 15-pound steel mass falling 20 inches to strike an anvil deeper into a hand-augured hole. Count the number of blows needed to drive the anvil down 6-inch increments.

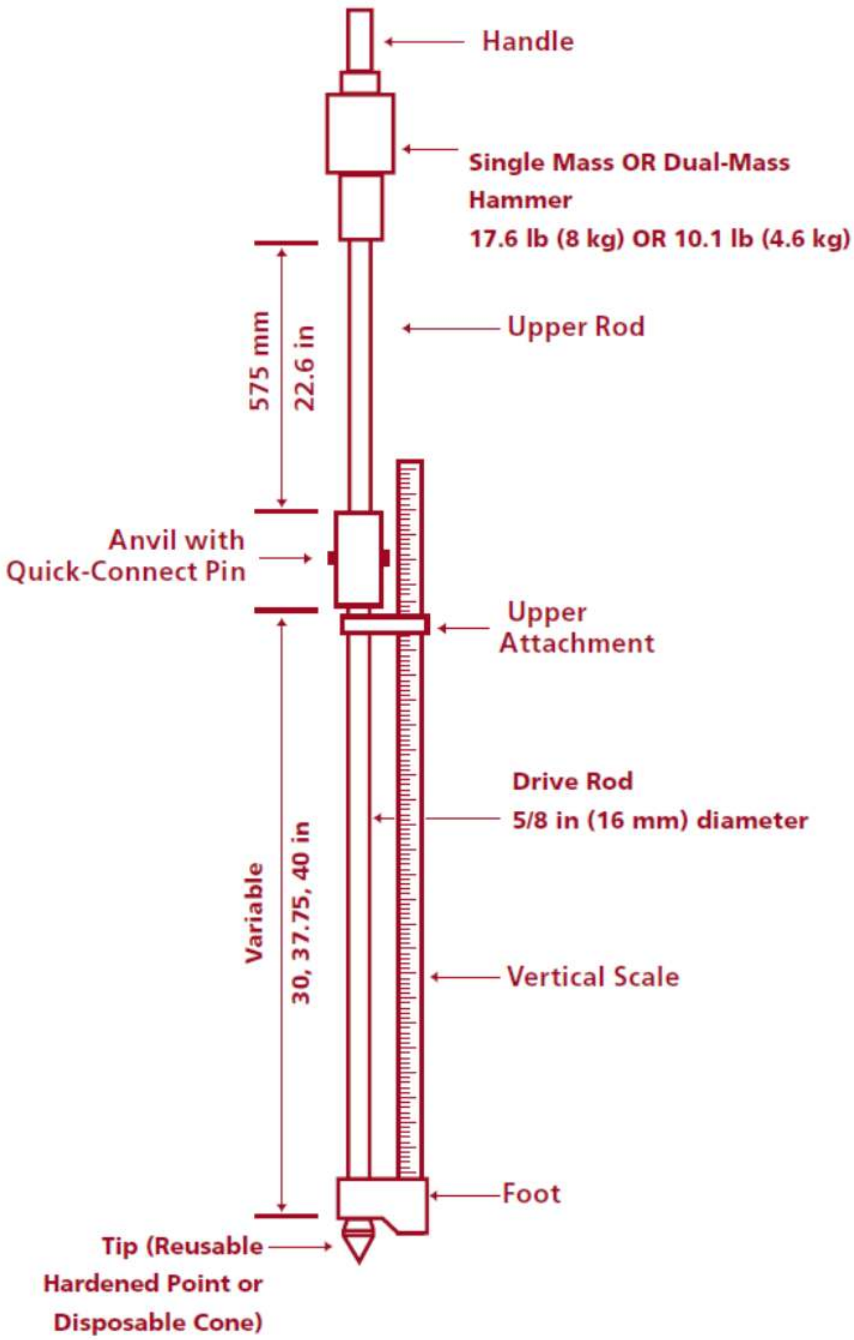

Figure 4.2.1: Dynamic cone penetrometer (Tensar)

4.3: Double Ring Infiltrometer

Infiltration rates impact irrigation, drainage, and how well water flows to crop roots. Infiltration measurements are used to predict erosion and determine soil health. In urban settings, stormwater systems and landfills need soil infiltration measurements to maximize or minimize water movement in the soil.

In a double ring infiltrometer, the inner ring is used as the measurement of the speed of water entering the ground. The outer ring is to ensure the water in the inner ring travels only vertically.

 Video Transcript

CHAPTER OVERVIEW

5: Groundwater

5.1: Groundwater Well Construction

Groundwater sampling wells are bored or drilled into the ground. They consist of an outer casing to create support for the sampling pipe. The sampling pipe itself has screened openings at the bottom, below the water table, to allow water to flow in but to block any sediment. Grouting is often used between the well pipe and the outer casing to create a waterproof barrier that prevents groundwater contamination.

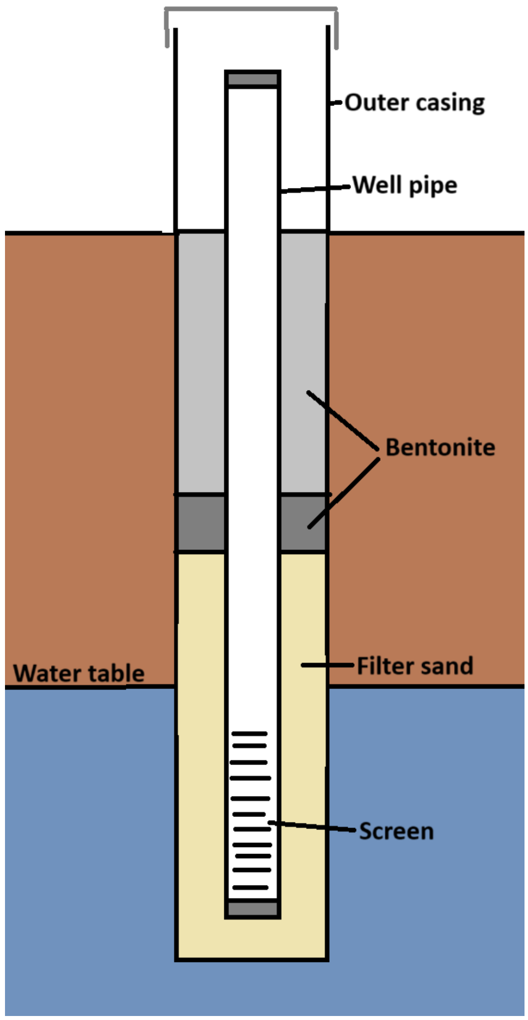

Figure 5.1.1: Diagram of groundwater sampling well.

5.2: Groundwater Well Sampling Techniques

Purpose

Beneath our feet, an underground water world exists. This groundwater, stored in aquifers, flows and rises and falls much like our surface water sources. How can we tell what's happening underground? We will perform groundwater well sampling. We can assess the following in the field:

- The depth of the water table
- The well recharge rate
- Groundwater flow direction
- Water quality

Equipment

- PPE
- Basic tool kit
- Sample bottles and sample sheet
- Electronic water level meter
- Disposable plastic water bailers
- String or cording
- Bucket(s)
- Field testing kits

Procedures

Measuring the groundwater depth

1. Make sure the electronic water level meter/probe is turned on and functioning.
2. Identify the point on the well casing used for measuring depth.
3. Lower the probe into the well until the reel indicates you have reached water.
4. Slowly raise and lower the probe to reach an accurate reading.

Measuring the well volume

1. Use the electronic water level meter/probe to measure to the bottom of the well.
2. Subtract the water table level from the well depth level to find the depth of the water column.
3. Use a ruler to measure the inside diameter of the casing.
4. Calculate the volume.

$$V = \frac{\pi D^2}{4} \times h$$

Purging the well

Purging the well removes stagnant water and gives us a fresh, representative sample.

1. Calculate the amount of water in four well volumes:

$$V = \frac{\pi D^2}{4} \times h \times 4 \times 7.48 \frac{gal}{ft^3}$$

1. Mark the cord so you can quickly lower the bailer to the water, then slowly let the bailer fill.
2. Let the bailer fill with water, then pull it up and pour the water into a calibrated bucket. Use a windmill motion with your arms to wind the cord.
3. Purge the well of four well volumes, or purge the well dry twice.
4. Allow the well to recover before sampling.

Sampling the well

1. Have sample containers ready and labeled. Only open one container at a time to avoid contamination.
    - If samples need preserving, add them within 15 minutes of sampling.
    - If preservative is already in the sample bottle, do not rinse the bottle before filling.
2. Use the bailer to pull up a sample of water.
3. Use a bottom-emptying bailer to decant your sample into a container.

Bailing Demonstration

Video Transcript

CHAPTER OVERVIEW

6: Total Maximum Daily Load

6.1: Watershed Health

Background

Scientists can evaluate impaired surface waters by performing a Total Maximum Daily Load (TMDL) analysis. A TMDL is the calculated amount of pollutant a water body can receive while still meeting water quality standards. This analysis, paired with restoration plans and projects, can address surface water quality concerns.

> **Total Maximum Daily Load (TMDL)**
>
> TMDL = Wasteload Allocation + Load Allocation + Margin of Safety
>
> Where:
>
> Wasteload Allocation is the total allowable pollutant load from all point sources.
>
> Load Allocation is the allowable pollutant load from non-point sources and natural sources.
>
> Margin of Safety accounts for uncertainty in the analysis.

TMDLs are a tool to support the Clean Water Act's goal of restoring impaired waters to meet water quality standards. The TMDL is approved by both the State of Wisconsin and the US Environmental Protection Agency.

6.2: Lower Fox River Basin

Lower Fox River Basin TMDL Program Background

The Lower Fox River Basin (LFRB) is approximately 640 square miles and extends from the outlet of Lake Winnebago to Green Bay. The TMDL program relies on citizen volunteers to collect samples from 20 unique monitoring sites. The two main goals of the project are:

1. Increase public awareness and involvement in water quality issues by engaging the public in citizen science
2. The collection of reliable surface water quality data to assess long-term water quality trends/successes

The program monitors several parameters, including:

- Total phosphorus
- Total suspended solids
- Total nitrogen
- Stream flow
- Water transparency

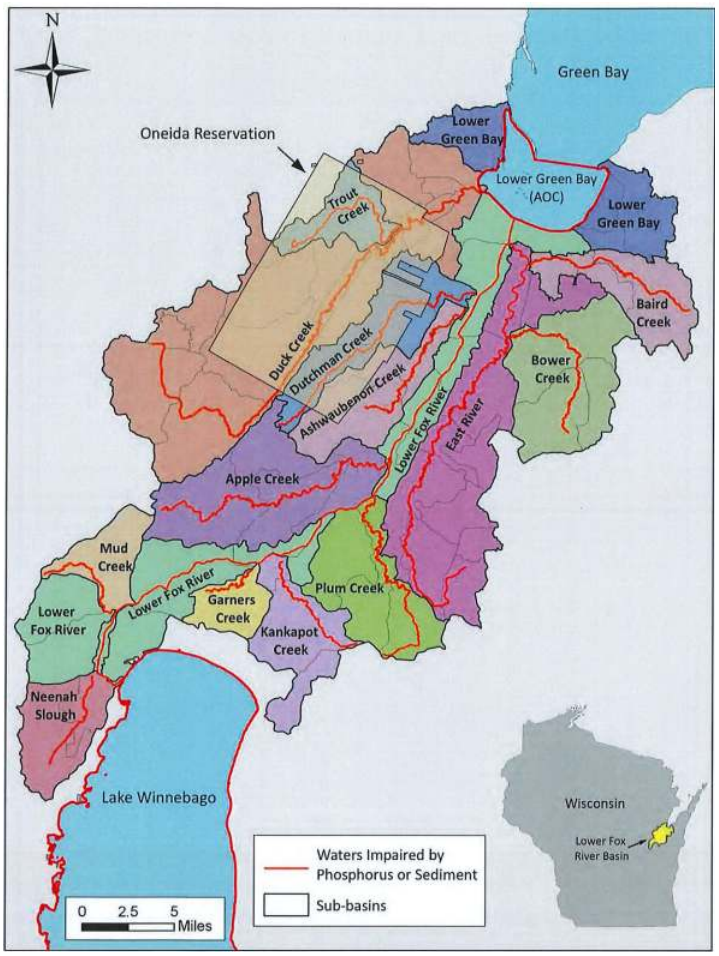

Figure 6.2.1: Sub-basins in the Lower Fox River Basin. (Image from Wisconsin Department of Natural Resources).

Program Progress

Volunteer monitoring began in 2015, with additional sites being added in 2018 and 2020. As the sampling program ages, trends in data become more clear. The Department of Natural Resources has formed the following key takeaways from the monitoring data analyzed from 2015 to 2023:

- Total phosphorus concentrations have decreased across many monitoring sites, with two sites achieving data below target levels.
- Dissolved reactive phosphorus concentrations remain high.
- Total phosphorus and total suspended solids concentrations correlate.
- Total suspended solids concentrations correlate with transparency tube readings.
- Rainfall contributes to the nutrient loading measured at sample sites.

6.3: Volunteer Sampling Instructions

When sampling, it is important to choose a location where water is flowing. If you need to enter the stream, enter at a point downstream and then walk up to the sampling location. This ensures that sediment disturbed by your movement is not collected. If entering the stream is not possible, a sampling pole can be used to grab a sample.

Transparency Testing

Turbidity measures the amount of suspended particles and the color of the water. Sources of turbidity can include soil erosion, runoff, algae, and bottom sediments. Turbidity blocks light from traveling through the water and can have an impact on photosynthesis and temperature. Turbidity can also impact fish that hunt by sight.

Turbidity tubes use a visual test. When collecting water, you may plunge the tube into the water or use a bucket or sampling pole to transfer water from the stream to the tube. Once the tube is filled, block direct sunlight, then look straight down from the top of the tube. You are looking for a black and white disc at the bottom.

- If you **can** see the disc, record the entire tube length on the form
- If you **cannot** see the disc at the bottom, release some of the water using the valve at the bottom of the tube until you can just barely see the disc. Record the height of the water in the tube.

The following video shows the Learn Water Action Volunteers' Stream Monitoring protocols to monitor transparency in streams.

Video Transcript

This season, the DNR is collecting visual information. Photos of the overall sampling area and a photo of a water sample in a mason jar should be emailed to the coordinator.

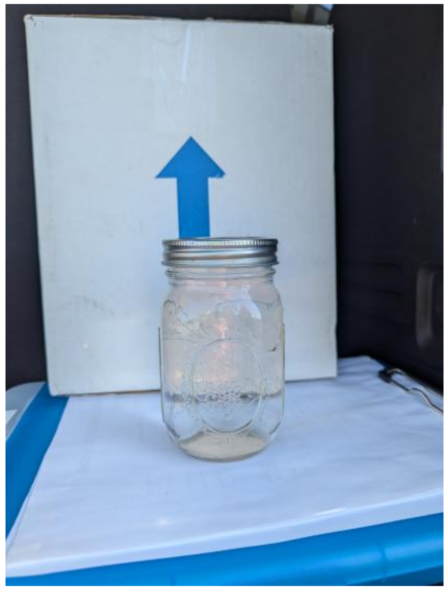

Figure 6.3.1: Water sample collected in a mason jar for visual information submission

Water Sampling

Sample bottles should be rinsed with sample water prior to filling. The sample should also be taken below the surface to collect a representative sample and avoid collecting contaminants.

- Open the bottle without touching the inside.
- Hold the bottle facing down, then submerge it about six inches into the water.
- Tip the bottle up to release the air and collect the water.
- Since you are rinsing the bottle three times, dump the collected water and repeat.
- Once the official sample is collected, cap the bottle.
- Some samples may require filtering or the addition of an acid preservative.

Packing and Shipping

Sample bottles are placed in a one-gallon Ziploc bag. Fill the remaining space in the cooler with ice contained in its own Ziploc bag. Double check that all fields on the lab slip have been filled out, then store the lab slip in a separate Ziploc bag and place it on top of the cooler within the box. Samples will be shipped to the Wisconsin State Lab of Hygiene.

Video transcript

CHAPTER OVERVIEW

7: Stormwater

7.1: Stormwater Runoff Management

Background

Urban stormwater runoff contains pollutants from roads, parking lots, construction sites, industrial storage yards, and lawns. Storm sewers make up an underground collection system to direct water away from homes, businesses, and roads. Storm sewers can discharge into stream beds, rivers, or lakes.

Many types of pollution can be found in urban stormwater, and this pollution can affect local streams and lakes. Contaminated stormwater is the most significant contributor of pollutants to Wisconsin's urban waters, affecting fish, wildlife, native vegetation, and drinking water sources.

- Garbage
- Oil and grease
- Gasoline
- Sediment from construction sites and urban runoff
- Metal flakes from rusting vehicles and brakes
- Road salt
- Lawn pesticides
- Agricultural herbicides
- Heavy metals from roof shingles
- Pet waste
- Leaves
- Grass clippings
- Bacteria
- Nutrients such as phosphorus and nitrogen
- Illicit discharges such as paint, cleaning solution products, and used motor oil.

Regulation

Directed by the Clean Water Act, the Wisconsin DNR developed the Wisconsin Pollutant Discharge Elimination System (WPDES) Storm Water Discharge Permit Program. This program regulates stormwater from:

- Construction sites
- Industrial facilities
- Municipal storm sewers

Municipal Storm Sewers

Municipal storm sewers in Wisconsin are required to provide information and education to the public, create and enforce local ordinances, and control total suspended solids (TSS).

Citizens can reduce the water volume to the storm sewer system in several ways:

- Diverting roof runoff
- Maintaining a healthy lawn
- Considering porous pavement or pavers for driveways and patios
- Building rain gardens

Many citizens do not consider the storm sewer when performing everyday activities. Washing a car in the driveway can introduce soaps and detergents into the system. Draining a pool can send chlorine into natural waterways. Citizens can reduce pollution to the storm sewer system in several ways:

- Cleaning up pet waste
- Keep leaves and yard waste out of the street
- Avoid overuse of pesticides and fertilizers

- Limit sidewalk and driveway salt
- Clean up chemical spills on hard surfaces

7.2: Erosion Control Methods

There is a high risk of sediment pollution in stormwater on construction sites due to the removal of vegetation and exposure of sediment. Best Management Practices (BMPs) for erosion control have been developed to limit sediment runoff from construction sites. BMPs can be divided into two categories:

- **Erosion Control** directly protects the disturbed soil surface from erosion
- **Sediment Control** aids in the removal of sediments from stormwater by using barriers or containment.

Erosion Control Methods

- Apply water to bare surfaces to reduce erosion from wind.
- Add erosion mats made of straw, wood, or other fibers/materials to protect soil surfaces.
- Use mulch to create a layer above the soil surface to protect it from rain impact and overland flow.
- Seed to establish temporary or permanent vegetation.
- Employ construction site diversions that are temporary berms or channels to divert runoff away from a construction site.

Figure 7.2.1: Example of bare sediment compared to a straw and fiber erosion mat coverage.

Figure 7.2.2: Erosion mat coverage of reseeded construction area near a stormwater pond.

Sediment Control Methods

- Add water-soluble substances to waterways to remove suspended solids within sediment control structures.
- Use vegetative buffers that can trap sediment within runoff by slowing down the flow velocity.
- Create a bale buffer made of anchored straw bales to intercept sediment-laden stormwater flow.
- Add a silt fence. This is an entrenched geotextile fabric to intercept sediment-laden stormwater flow.
- Employ trackout control practices to aid in removing dirt and debris from vehicles exiting the construction site.
- Use storm drain inlet protection to temporarily block sediment with fabric.
- Create a ditch check. This is a temporary dam made of straw bales located in ditches or swales susceptible to site runoff.
- Employ a sediment trap/basing. This is a temporary or permanent excavated area to hold water for a sufficient time for sediment to settle out.

Figure 7.2.3: Silt fence installation along the construction road and a course gravel trackout pad to control sediment.

Figure 7.2.4: Storm drain inlet protection.

Index

Glossary

Sample Word 1 | Sample Definition 1

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Overview

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